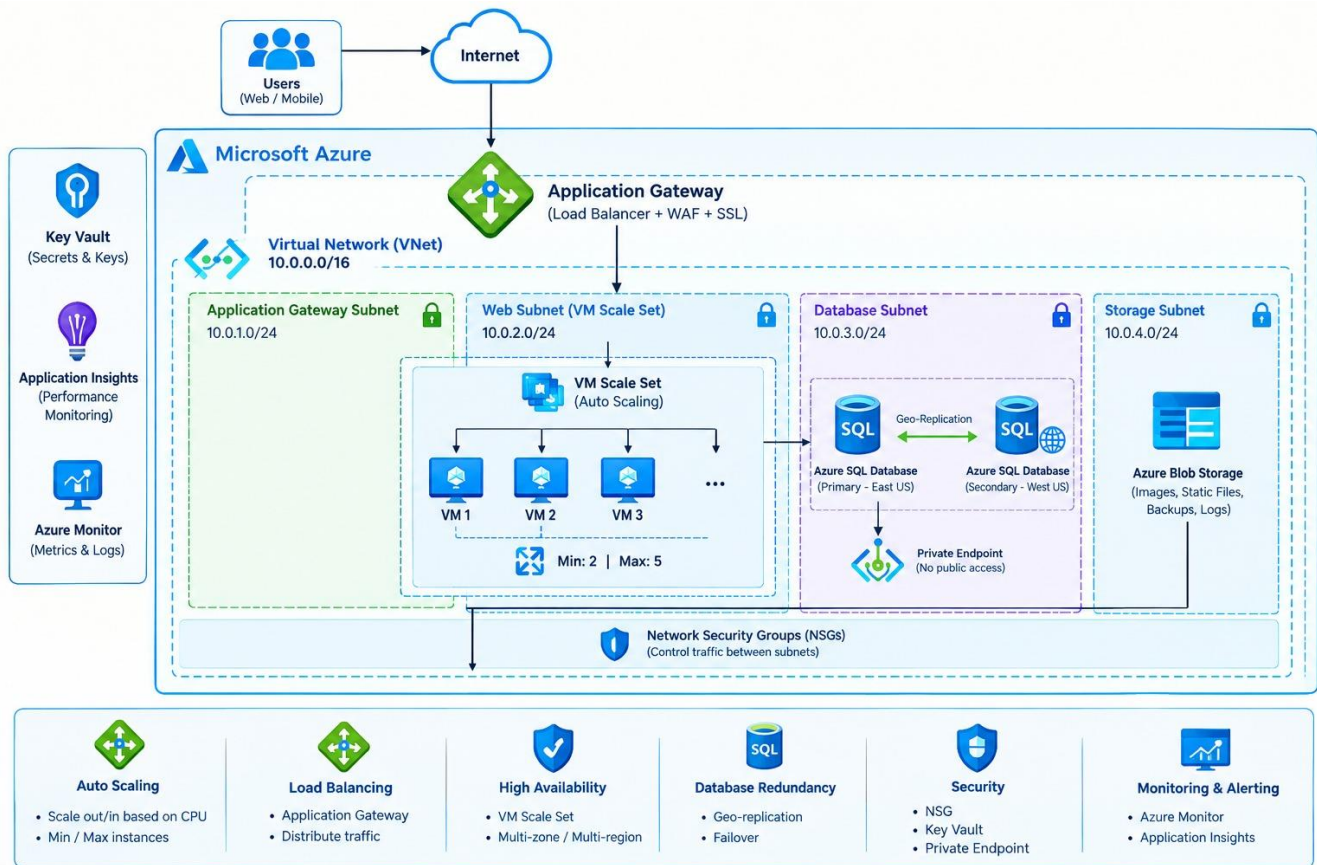


Azure Cloud Infrastructure Task



1. Introduction

This task focuses on designing and implementing a secure, scalable, and highly available web application infrastructure using Microsoft Azure. The architecture is designed to support web users, automatic scaling, secure database access, data storage, monitoring, and disaster recovery.

2. Architecture Overview

Users access the application through the Internet using web or mobile devices. Traffic is received by Azure Application Gateway, which provides load balancing, Web Application Firewall (WAF), and SSL/TLS termination. The Application Gateway forwards requests to a Virtual Machine Scale Set (VMSS) located inside the Web Subnet.

3. Azure Network Design

The infrastructure uses an Azure Virtual Network (VNet) with the address range 10.0.0.0/16. The VNet is divided into the following subnets:

- Application Gateway Subnet: 10.0.1.0/24
- Web Subnet: 10.0.2.0/24
- Database Subnet: 10.0.3.0/24
- Storage Subnet: 10.0.4.0/24

Network Security Groups (NSGs) will be used to control and restrict traffic between the different subnets.

4. Application Layer

The web application will run on an Azure Virtual Machine Scale Set. The VMSS will support automatic scaling based on workload, with a minimum of 2 instances and a maximum of 5 instances. This helps maintain application performance during high traffic while reducing resource usage during low traffic.

5. Database and Storage

The application will use Azure SQL Database for relational data. The architecture includes a primary database in East US and a secondary database in West US with geo-replication for redundancy and disaster recovery. A Private Endpoint will be used to prevent public access to the database.

Azure Blob Storage will be used for storing images, static files, backups, and logs.

6. Security and Monitoring

Security will be implemented using several Azure services:

- Azure Key Vault for storing secrets and keys.
- Network Security Groups for network traffic control.
- Private Endpoint for secure database connectivity.
- Application Gateway WAF for web application protection.

Azure Monitor and Application Insights will be used to track application performance, metrics, logs, and alerts.

7. Expected Outcome

The planned architecture aims to provide:

- Scalability through Virtual Machine Scale Sets.
- High availability through multiple instances and regions.
- Load balancing using Application Gateway.
- Database redundancy through geo-replication.
- Strong security using WAF, NSGs, Key Vault, and Private Endpoints.
- Monitoring and alerting using Azure Monitor and Application Insights.

8. Conclusion

The task will involve implementing a secure, scalable, and highly available Azure infrastructure based on the proposed architecture. The main goal is to build an environment that can handle changing workloads while maintaining performance, security, and reliability.